

Safer food and pharmaceuticals using IoT

Tell us your challenge. We're here to help.

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Google Cloud Platform

- Doubles number of customer service calls per period
- Adds two countries to the service footprint
- Improves customer service d

When your job is to protect the food and pharmaceuticals we take from contamination, protecting businesses' reputations for hygiene is a top priority. That's why we've developed its Connected service.

PestConnect is a remote monitoring system that adds wireless connectivity to remote sensors. Technicians via the internet when the system is activated. It's the smartest mouse trap.

The system was developed in house at Rentokil

ABOUT RENTOKIL



About Rentokil

Rentokil Initial is one of the largest business services companies in the world, providing pest control, hygiene, and workwear services in over 60 countries.

Industries: Other

Location: United Kingdom

[Dataflow](https://cloud.google.com/dataflow/) (<https://cloud.google.com/dataflow/>)

Initial (RI) which is the global leader in hygiene and pest control services, also offers HygieneConnect, a solution that calculates the percentage of occupants who wash their hands, those who don't and displays the results in real time on a monitor in the venue (anonymously, of course). This helps increase handwashing compliance. And, in case you're wondering, the difference between the handwashing compliance of men and women!

Both services seemed promising. With a customer base globally, RI needed a solution with scalability, speed, and data stream processing technology that only a cloud solution could provide. Step forward PA Consulting Group (PA), using Google Cloud Platform, designed and delivered a solution to help Rentokil Initial to achieve its growth ambitions.

About PA Consulting Group

PA Consulting Group is a global consulting, technology, and innovation firm with over 2,600 people.

Group

BigQuery (<https://cloud.google.com/bigquery/>)

PA Consulting Group is a global

App Engine (<https://cloud.google.com/appengine/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Cloud Datastore

(<https://cloud.google.com/datastore/>)

PA Consulting

Dataflow (<https://cloud.google.com/dataflow/>)

An IoT solution for nearly infinite growth

RI's on-premises platform simply couldn't handle the amount of streaming data that true growth demanded. The sensor-equipped devices (rodent traps and soap dispensers) send data wirelessly to control panels, which process the information. That information is made immediately available to customers on the myRentokil portal (which includes features such as trend reporting, recommendations, charts, and analytical tools). RI's data center could

only handle messages generated by about 1,500 control panels, which roughly correlates to 1,500 customers. The company wanted to grow the business to 10,000 customers in the near term. That called for a cloud solution that could ingest data continuously at a massive scale and securely. After considering several IOT cloud-service providers and delivery partners, RI chose Google Cloud Platform and PA, a global consulting, technology, and innovation firm and award-winning Google partner. PA's business intelligence and analytics expertise, combined with its digital business capabilities, made it an ideal partner to enhance growth while addressing Rentokil's underlying technology and connectivity challenges.

High performance at lower cost

App Engine provided the necessary scalability, durability, and resilience to scale with demand and handle unexpected spikes in traffic. In addition, App Engine enabled an incredibly fast 12-week development of RI's connection technology. During testing, after simulating 100,000 device data packets—ten times more than RI's end-of-year goal—the error rate was 0%, with an average response time to devices of under 200 ms. This extremely low latency allowed RI's IoT devices to quickly free up their memory and processing capacity, and to focus on the task at hand.

The IoT: Taking data from small to big

An IoT solution, combined with new kinds of device technology, makes it easy and economical to mine information from any physical item and place—in RI's case, rodent traps and soap dispensers. Core to such a solution is a solid big-data toolkit, and PA was able to deliver it on Google's platform.

RI required a low-cost solution to store all the data coming from its devices and also to query the current state of devices. Cloud Datastore delivered on functionality as well as a competitive pricing model. In addition, its high availability and consistency enabled an audit of all device data that the RI staff could easily view. Cloud Datastore's NoSQL structure could be queried easily through the cloud dashboard with a minimum of training. Finally, Cloud Datastore's encryption-at-rest security meant that credentials for both devices and external systems could be saved in Cloud Datastore for authentication purposes.

Cloud Pub/Sub's push implementation was used to manage unexpected high loads of device messages. Cloud Pub/Sub enables secure and highly available communication between independently written applications, allowing developers to quickly integrate systems hosted on Google Cloud Platform and externally. Cloud Pub/Sub's ability to store messages in a queue offered a way to ensure that RI lost no device data. The push implementation also offered near real-time performance.

Cloud Dataflow served as a pipeline through which messages from devices could be passed seamlessly from Cloud Pub/Sub into BigQuery and broken down into an easily readable form. The data

could then be sent to the correct table in BigQuery with very low latency. BigQuery's ability to analyze and partition the device data into monthly tables made it easy for Qlik Sense (a browser-based tool used to create interactive visualizations from multiple data sources) to create KPIs from the data consumed.

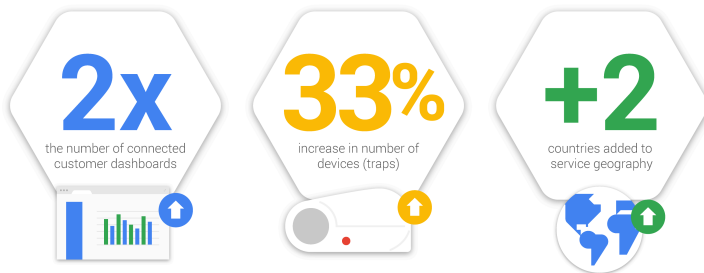
"Due to the collaboration between Rentokil Initial, Google Cloud Platform, and PA's agile development experts, we were able to deliver a real IoT solution in just three months across multiple countries," said Dan Rossner, digital solutions expert at PA. "We're proud to have helped grow Rentokil Initial's connected business."

Real-time, actionable insights

With large-scale data streaming and processing capabilities, the RI team can now analyze millions of rows of data in seconds, providing a real-time snapshot of how its IoT infrastructure is performing. Having this vast collection of accessible data at its fingertips allows RI to derive new insights—like understanding when devices need attention and providing near-time information on battery levels, connectivity, and activations—that lead to better business decisions.

Since Rentokil Initial completed the project with PA, it has doubled the number of connected customer dashboards, while the number of devices has increased by a third, and launched its Connected services in 12 countries. This means it is now positioned to meet its growth targets for 2016 and

well into the future, providing high-quality customer insight to maintain its position as a global market leader.



"Moving to Google Cloud Platform is a great step forward for our platform and provides a strong foundation for the future," remarks Paul Donegan, Corporate IT Director of Rentokil Initial. "The scalable IoT platform will enable us to accelerate our innovation in both the pest control and hygiene sectors."